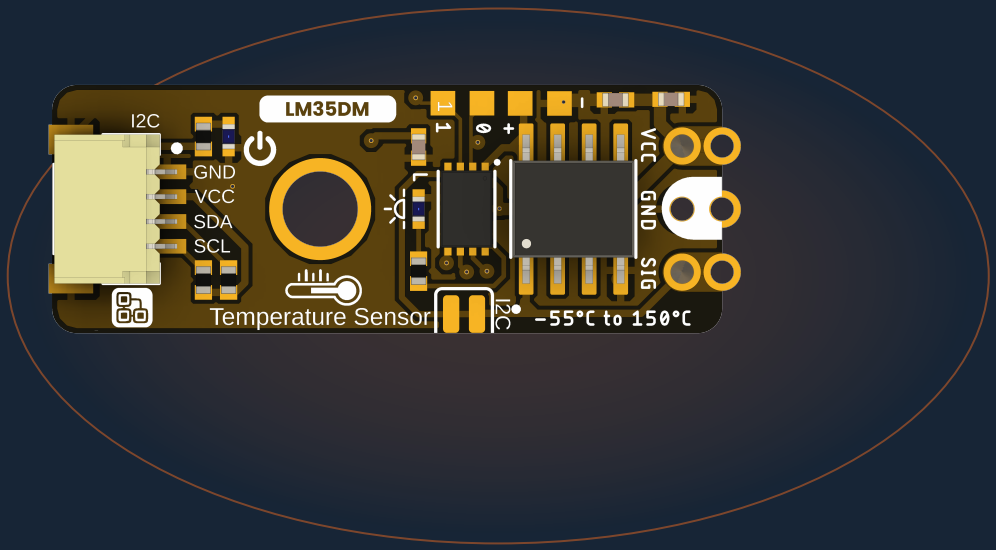
PRODUCT REFERENCE MANUAL

UNIT ELECTRONICS · DEVLAB ECOSYSTEM

DevLab I2C LM35 Temperature Sensor

MPN: UE0169

Version 1.1.0



I2C temperature sensor module based on the LM35.

TEMPERATURE | I2C | DEVLAB ECOSYSTEM

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Description

The DevLab I2C LM35 Temperature Sensor (MPN **UE0169**) is an I2C temperature sensing module in the UNIT Electronics DevLab family. The product-specific board views and any released drawings are collected under `hardware/`.

Applications

- Temperature sensing in an I2C-based prototype.
- Evaluation of the module with a compatible host controller.

Available Documentation

A product-specific schematic was not present in the selected source package. See Chapter 8 for the available source files. Confirm the electrical interface against the product-specific pinout or schematic before connecting the board.

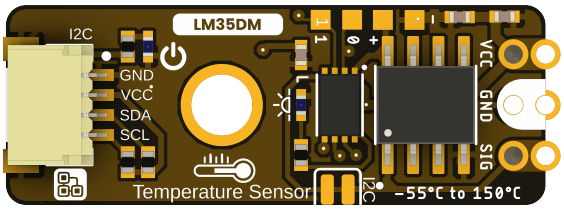
1 The Board

1.1 Board Identification

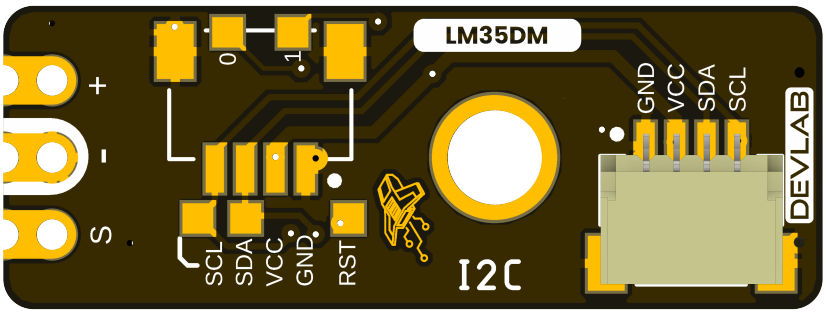
Item	Identification
Product	DevLab I2C LM35 Temperature Sensor
Manufacturer Part Number (MPN)	UE0169
Function	Temperature sensing

1.2 Board Views

The following views come from the selected UE0169 source package. They show component placement and board orientation; use a dimensioned drawing for mechanical tolerances.



DevLab I2C LM35 Temperature Sensor top view.



DevLab I2C LM35 Temperature Sensor bottom view.

2 Ratings

The selected UE0169 source package does not establish a complete set of electrical ratings in this reference. Do not use values from another DevLab module as substitutes.

2.1 Electrical Limits

Supply range, I/O voltage limits, and current consumption require confirmation from the product-specific schematic and component data before integration.

2.2 Interface Characteristics

Confirm the I2C address, bus speed, pull-up arrangement, and connector orientation with the released UE0169 pinout or firmware documentation. An I2C scan only identifies responding addresses; it does not verify the device protocol.

3 Functional Overview

The DevLab I2C LM35 Temperature Sensor provides temperature sensing for an I2C-based system. The supplied board views identify the physical module, while the product-specific schematic and firmware documentation are needed to establish its internal signal path and command set.

3.1 Integration Boundary

Use the available UE0169 pinout, schematic, and firmware release to determine power, connector mapping, address, and commands. Until these are confirmed, treat the generic I2C scanner in `software/` as a bus diagnostic only.

4 Connectors and Pinout

A released pinout drawing for the UE0169 I2C LM35 Temperature Sensor was not present in the selected source package. The top and bottom views in Section 6 show the visible board labels, but they do not establish connector numbering, voltage limits, or signal direction. Verify these details against the product-specific schematic before connecting a cable or designing a carrier.

5 Board Operation

5.1 Before Connecting

Check the product-specific UE0169 pinout or schematic for power, ground, SDA, and SCL. The board views alone do not establish voltage limits or connector numbering.

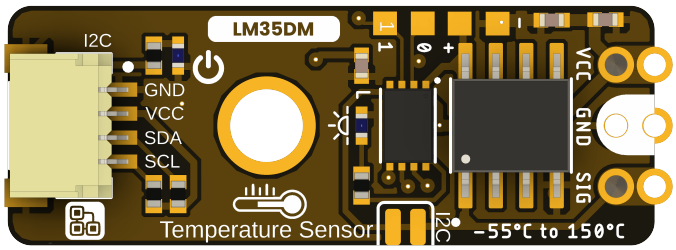
5.2 I2C Bus Check

After confirming the wiring and supply limits, use the generic Arduino or MicroPython I2C scanner in `software/examples/i2c/`. Set the host SDA and SCL pins for your controller. A responding address confirms only that a device acknowledged on the bus; it does not identify the module or validate its measurement protocol.

6 Mechanical Information

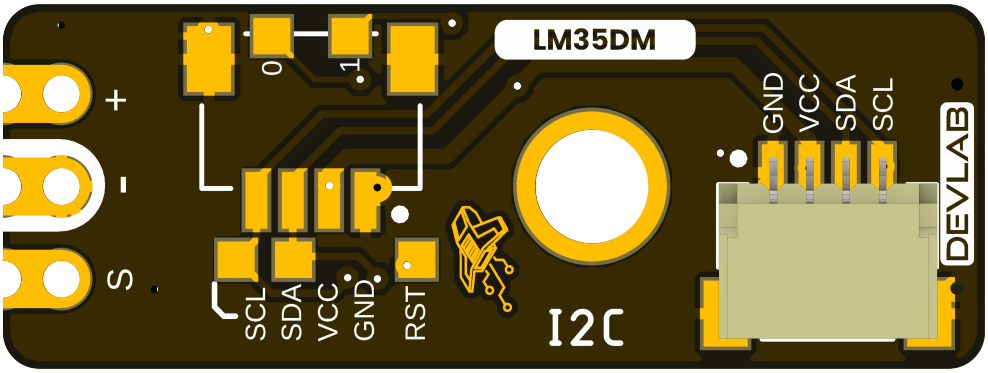
The supplied board views identify the component and underside layouts of the UE0169 I2C LM35 Temperature Sensor. Use the labeled drawings for fit checks; do not infer dimensions by scaling a render.

6.1 Top View



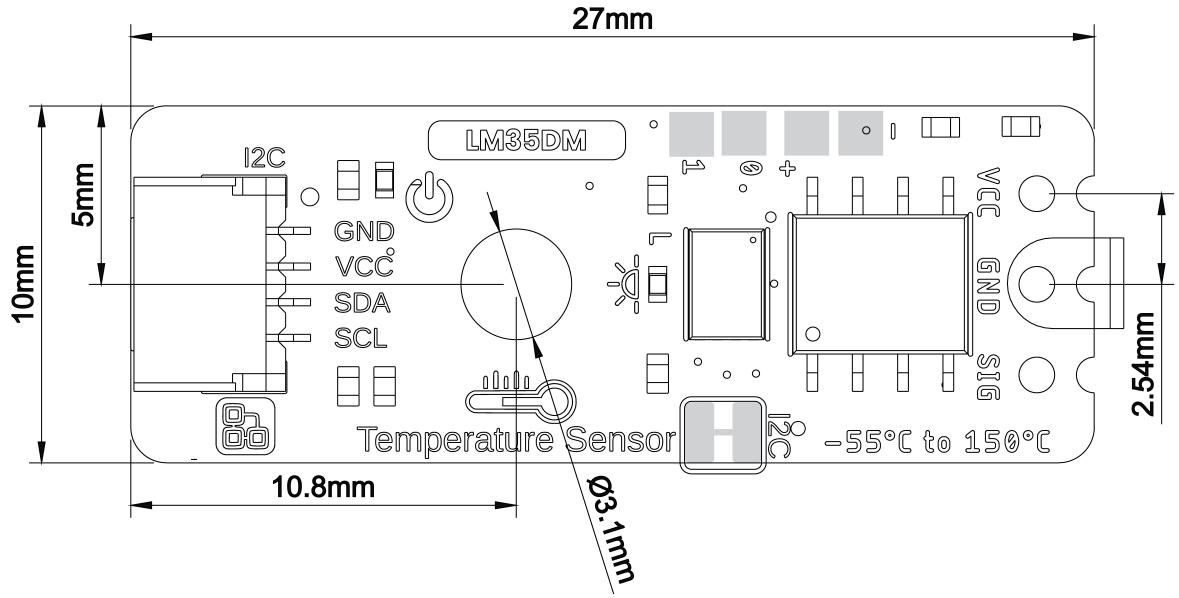
UE0169 I2C LM35 Temperature Sensor top view.

6.2 Bottom View



UE0169 I2C LM35 Temperature Sensor bottom view.

6.3 Board Dimensions



Board Dimensions

UE0169 I2C LM35 Temperature Sensor dimension drawing.

Use the dimensions printed on the supplied drawing. PCB thickness, component heights, connector clearance, and manufacturing tolerances require separate verification.

7 Company Information

Item	Information
Company / brand	UNIT Electronics
Board ecosystem	DevLab
Product family	Atom
Website	UNIT Electronics
Address	Salvador 19, Cuauhtémoc, 06000 Mexico City, CDMX, Mexico

The product repository and controlled technical references are listed in the next chapter.

8 Reference Documentation

Reference	Location
Product repository	UE0169 I2C LM35 Temperature Sensor
Top view	3D board view
Bottom view	3D board view
Board dimensions	Dimension drawing
Board dimensions PDF	PDF
Board topology	Topology drawing

A product-specific pinout drawing was not included in the selected source package.

9 Appendix

9.1 Schematic Status

A product-specific schematic was not included in the selected source package.

9.2 Document Control

Item	Value
Product	DevLab I2C LM35 Temperature Sensor
MPN	UE0169
Source	UNIT Electronics selected product source package

9.3 Data Requiring Confirmation

Electrical limits, I2C commands, and mechanical tolerances must be verified against product-specific released documentation.